

GridSentinel AI

v2.0 — Smart Grid Intelligence Platform for BESCOM

Project	GridSentinel AI — Electricity Theft Detection & Grid Stress Monitoring
Client	Bangalore Electricity Supply Company (BESCOM)
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1. Executive Summary

GridSentinel AI v2.0 is a production-grade smart grid intelligence platform developed for the Bangalore Electricity Supply Company (BESCOM). The system addresses two critical operational challenges in modern power distribution: **electricity theft detection** at the meter level and **transformer stress monitoring** at the distribution transformer (DT) level.

The platform ingests hourly smart meter telemetry for up to 1,000 meters across 21 distribution transformers, applies a 7-signal anomaly scoring engine (CASS), an 8-signal grid stress index (GSI), a Bi-LSTM quantile demand forecaster, and an XGBoost theft classifier. Results are surfaced through a real-time React dashboard with AI-powered natural language querying via Ollama/LLaMA 3.2.

Outcome	Description	Scale
CASS Score	Comparative Anomaly Signal Score per meter — theft suspicion index	0 to 100
GSI Score	Grid Stress Index per transformer — operational risk level	0 to 100
GSS Core	Composite system score combining theft, forecast and economic quality	0.0 to 1.0
GSS Final	Extended GSS including calibration, energy balance, detection latency	0.0 to 1.0
Economic KPIs	Revenue protection vs investigation cost in Indian Rupees (INR)	INR

2. System Architecture & Workflow

2.1 High-Level Block Diagram

The following diagram illustrates the end-to-end data flow from raw CSV ingestion through to the frontend dashboard.

```

[CSV / Smart Meter Data] (672,000 rows x 14 columns)
|
v
+-----+
| Ingestion Layer | Column mapping, type casting, 3-phase synthesis,
| (loader.py) | billed_kWh derivation, schema validation
+-----+
|
+-----+-----+
| |
v v
meter_df dt_df
(672k rows) (14k rows)
| |
v v
+-----+ +-----+
| Feature Store | | Forecast Features |
| (7 CASS signals | | (13 LSTM features |
| + 7 engineered) | | per DT per hour) |
+-----+ +-----+
| |
v v
+-----+ +-----+
| XGBoost Theft | | Bi-LSTM Demand |
| Detector | | Forecaster |
| (14 features in) | | (Q5, Q95 out) |
+-----+ +-----+
| |
v v
+-----+ +-----+
| CASS Scorer | | GSI Scorer |
| (weighted sigmoid | | (8 signals weighted |
| per meter) | | per transformer) |
+-----+ +-----+
| |
+-----+ +-----+
| |
v v
+-----+
| GSS Compositor |
| (7 sub-scores -> |
| GSS Core/Final) |
+-----+
|
v
+-----+
| FastAPI REST API |
| (SQLite + JWT) |
+-----+
|
v
+-----+
| React Dashboard |
| Vite + Recharts |
| Ollama LLM Chat |
+-----+

```

Figure 1: GridSentinel AI end-to-end data flow

2.2 Pipeline Steps (18 Steps)

The **GridSentinelPipeline.run()** method executes the following ordered steps:

Step	Name	Description
1	Schema Validation	Validates column presence, types and value ranges for meter_df and dt_df using validator.py
2	Resample & Align	Sorts by meter_id/dt_id + timestamp, removes duplicate (meter, timestamp) pairs
3	Meter Feature Building	Computes DTW clusters via tslearn KMeans on hourly profiles, then calculates all 7 CASS signals and 7 engineered features for each meter
4	Train/Test Split	Time-based 80/20 split on sorted unique timestamps — preserves temporal ordering
5	TheftDetector Training	Trains XGBoost classifier on training-set feature matrix with scale_pos_weight for class imbalance; saves to models/theft_detector.json
6	CASS Score Computation	Applies weighted sigmoid formula to 7 signals per meter using trained model probabilities where available
7	GSI Signal Computation	Calls compute_thermal_hours() to derive hours_above_80pct rolling sums per DT
8	DemandForecaster Training	Builds 13-feature forecast matrix per DT, trains Bi-LSTM with quantile loss on the first DT as representative; saves model + scaler
9	GSI Score per DT	For each transformer: runs demand forecast, extracts Q95, computes 8 GSI signals, derives GSI score
10	Classification Evaluation	Computes precision, recall, F1, FPR, MCC, ROC-AUC, PR-AUC, ECE on full feature matrix
11	Robustness Testing	Injects Gaussian noise (std=0.05) into features, measures F1 drop; $S_{Robust} = 1 - F1_{clean} - F1_{noisy} / F1_{clean}$
12	Temporal Stability	Counts isolated CASS spikes (score ≥ 60 surrounded by <35) as fraction of anomalies
13	Energy Consistency	$S_{Energy} = 1 - \text{sum_meter_kWh} - \text{sum_feeder_kWh} / \text{sum_feeder_kWh}$
14	Detection Latency	$S_{Delay} = \exp(-\text{latency_days} / 7)$; defaults to 1.0 when theft start timestamps unavailable
15	Calibration Score	$S_{Calib} = 1 - \text{ECE}$ (Expected Calibration Error from 10-bin reliability diagram)
16	Pareto Constraints	Checks: $FPR \leq 0.02$, $Recall \geq 0.85$, $MAPE \leq 0.07$; constraints_met flag penalises GSS by 50%
17	Economic Cost	$\text{cost} = FP_count \times 8500 + FN_count \times 5500 \times 3$; $S_{Econ} = 1 - \text{cost} / \text{baseline_cost}$

Step	Name	Description
18	GSS Computation	GSS Core and GSS Final via weighted sums of sub-scores; SHAP summary plot saved to results/shap_summary.png

3. Tools & Technologies

Tool / Library	Version	Role in GridSentinel
Python	3.11+	Primary runtime language for all backend components
FastAPI	0.110+	Async REST API framework — handles all HTTP endpoints
Uvicorn	0.29+	ASGI server hosting the FastAPI application
SQLAlchemy	2.0+	ORM for SQLite — User and AuditLog tables
SQLite	Built-in	Embedded database for users and audit logs
XGBoost	2.0+	Gradient-boosted tree classifier for electricity theft detection
PyTorch	2.2+	Deep learning framework powering the Bi-LSTM demand forecaster
tslearn	0.6+	Time-series DTW K-Means clustering for meter load profile grouping
SHAP	0.45+	SHapley Additive exPlanations for XGBoost model interpretability
LIME	0.2+	Local Interpretable Model-agnostic Explanations (available)
scikit-learn	1.4+	StandardScaler, KMeans fallback for clustering, metrics
scipy	1.12+	Shannon entropy calculation for the entropy CASS signal
pandas	2.1+	Tabular data manipulation, groupby, resampling, merges
numpy	1.26+	Array operations, DTW norm, sigmoid, polyfit for trend slope
pyarrow	15.0+	Parquet file I/O for alternative data loading path
React	19+	Frontend UI framework — component-based single-page application
Vite	latest	Frontend build tool with dev-server proxy to FastAPI
Recharts	3.8+	Chart library for DTW divergence and demand forecast visualisations
Axios	1.16+	HTTP client used in App.jsx for authenticated API calls
Lucide React	latest	Icon set used throughout the dashboard UI
python-jose	3.3+	JWT encoding/decoding for HS256 bearer token authentication
passlib + bcrypt	1.7+ / 4+	Secure password hashing for the user credential store
fpdf2	2.7+	PDF report generation for the /api/report/summary endpoint
ollama	0.4+	Python client for local LLaMA 3.2 inference via Ollama daemon
LLaMA 3.2	3.2	Local large language model for the AI chat assistant panel

Tool / Library	Version	Role in GridSentinel
matplotlib	3.7+	SHAP summary plot generation saved to results/shap_summary.png

4. Input Data Schema

The system accepts a single flat CSV file containing smart meter readings. Each row represents one meter reading at one timestamp. The loader maps CSV column names to internal schema names and derives additional columns automatically.

4.1 Required CSV Columns

CSV Column Name	Internal Name	Type	Description
Meter_ID	meter_id	String	Unique identifier for each smart meter (e.g. MTR_000001)
Transformer_ID	dt_id	String	Distribution transformer this meter is connected to (e.g. DT_1)
Timestamp	timestamp	Datetime	UTC datetime of the reading. Parsed with timezone awareness.
Active_Power (kWh)	kwh	Float	Energy consumed during this interval in kilowatt-hours
Voltage	voltage_r	Float	Primary phase (R-phase) voltage in volts
Current	current	Float	Current draw in amperes
Power_Factor	power_factor	Float	Ratio of real to apparent power. Range: 0.0 to 1.0
Is_Theft (0/1)	is_theft	Int	Ground truth theft label (1 = theft, 0 = normal). Used for model training and evaluation only.
Imputation_Flag	imputation_flag	Int	1 if this reading was imputed/estimated, 0 if measured directly
Imputation_Confidence	imputation_confidence	Float	Confidence score of the imputation model (0.0 to 1.0)

4.2 Optional CSV Columns

CSV Column Name	Internal Name	Description
Peer_Group_ID	peer_group_id	Pre-assigned peer group for comparative analysis. Used as passthrough metadata.
Theft_Type	theft_type	Category of theft (e.g. bypass, tamper). Metadata only — not used in model features.
GSI_Event_Type	gsi_event_type	Type of grid stress event associated with this reading. Metadata passthrough.

CSV Column Name	Internal Name	Description
Topology_Confidence	topology_confidence	Confidence score that the reported DT-meter topology is correct (0.0 to 1.0).

4.3 Derived Columns (Auto-generated by Loader)

Derived Column	Source	Description
voltage_y	voltage_r + noise	Y-phase voltage synthesised by adding uniform noise U(-2, +2) V to R-phase
voltage_b	voltage_r + noise	B-phase voltage synthesised by adding uniform noise U(-2, +2) V to R-phase
billed_kwh	Monthly kWh sum	Sum of kwh per meter per calendar month — represents the billed amount for billing ratio calculation
feeder_kwh	DT aggregate	Sum of all meter kwh under the same DT at the same timestamp — represents the feeder measurement
actual_kwh	feeder_kwh copy	Alias of feeder_kwh used as the LSTM forecast target
temperature_c	Diurnal formula	Synthetic: $28 + 4 \times \sin(2\pi \times \text{hour}/24)$. Ranges 24 to 32 degrees C.
solar_irradiance	Hour formula	Synthetic: $\max(0, 800 \times \sin(\pi \times (\text{hour}-6)/12))$ during daylight (6-18h), 0 otherwise
ev_density	Constant	Default 0.1 — EV penetration density proxy
capacity_kva	Constant per DT	Fixed at 500 kVA per transformer
age_years	DT index formula	Cycles through 5, 10, 15, 20 years based on transformer sort order

5. The 7 CASS Signals — Comparative Anomaly Signal Scoring

CASS (Comparative Anomaly Signal Scoring) is GridSentinel's primary theft detection framework. It computes seven complementary signals per meter, weights them, and applies a sigmoid transformation to produce a theft suspicion score from 0 to 100. Each signal is normalised to [0, 1] before weighting.

5.1 Signal Definitions

Signal	Weight	Formula / Logic	What It Detects
DTW Divergence	0.25	$\text{norm}(\text{meter_hourly_profile} - \text{cluster_centroid}) / \text{dtw_norm_denom}$ (default 3.0), clipped to [0,1]	Abnormal load shape relative to peer meters in the same DTW cluster. A thief often manipulates bypass circuits creating an atypical daily profile.
Voltage Stability	0.10	$\text{std}(\text{voltage_R}, \text{voltage_Y}, \text{voltage_B}) / 0.15$, clipped to [0,1]	Voltage irregularities across the three phases indicating meter tampering, loose connections, or bypass wiring that introduces noise.
Billing Ratio	0.20	$ 1 - \text{billed_kWh} / \text{actual_kWh} $, clipped to [0,1]	Discrepancy between billed consumption and measured consumption. Under-billing relative to actual usage is a strong theft indicator.
Entropy	0.10	$1 - (\text{Shannon entropy of 20-bin histogram} / \log(20))$	Unusually repetitive or predictable consumption patterns. Low entropy means the meter reports nearly the same value repeatedly — a sign of manipulation.
Night Load Anomaly	0.15	$\text{clip}((\text{night_avg} / \text{day_avg} - 1) / 2, 0, 1)$ Night = 22:00-05:59, Day = 06:00-21:59	Disproportionately high consumption at night relative to daytime. Theft often peaks at night to avoid detection. Signal is 0 when night $\leq$ day.
DT Balance Error	0.15	$ \text{sum_all_meters_kWh} - \text{feeder_kWh} / \text{feeder_kWh}$, clipped to [0,1]	Energy loss at the transformer level. If the sum of all meters is significantly less than the feeder reading, energy is being taken without metering.
Repeat Anomaly	0.05	Fraction of 96-interval windows where any Z-score > 3.0, divided by window_count (5)	Systematic repeating anomalous consumption windows. Tampered meters often produce regular spikes or flat segments that appear as statistical outliers repeatedly.

5.2 CASS Scoring Formula

The final CASS score is computed as follows:

```

raw_score = sum( weight[k] x signal[k] ) for k in {7 signals}
CASS = sigmoid( (raw_score - 0.35) x 8.0 ) x (1 - g_pv) x 100
g_pv = clip( solar_irradiance / 1000 x 0.3, 0.0, 0.3 )

```

The sigmoid shift (0.35) and scale (8.0) are calibrated so that a raw score below 0.35 maps to less than 50 CASS. The PV correction factor **g_{pv}** reduces the CASS score by up to 30% during high solar irradiance periods, preventing false alarms caused by grid-tied solar installations altering apparent consumption profiles.

5.3 CASS Label Thresholds

CASS Range	Label	Recommended Action
0 – 34	Normal	No action. Meter operating within expected parameters.
35 – 59	Watch	Monitor for trend. Flag for next billing cycle review.
60 – 79	Inspect	Schedule field inspection. Cross-check billing records.
80 – 100	Immediate	Immediate field intervention. Potential active theft in progress.

6. The 8 GSI Signals — Grid Stress Index

The GSI (Grid Stress Index) quantifies operational risk at the distribution transformer level. It combines eight physical and temporal signals to produce a stress score from 0 to 100. Unlike CASS (which is per meter), GSI is computed once per distribution transformer per time step using both measured telemetry and Bi-LSTM demand forecasts.

Signal	Weight	Formula / Logic	What It Measures
Load Quantile	0.30	$\text{q95_forecast_kWh} / (\text{capacity_kVA} \times \text{power_factor})$, clipped to [0,1]	The Q95 forecast load as a fraction of rated capacity. Values near 1.0 mean the transformer is near or above rated load — highest risk signal.
Temperature Derating	0.15	$\text{clip}((\text{temperature_C} - 32) \times 0.05, 0, 1)$	Thermal derating of transformer capacity above 32 degrees C. Every degree above 32 reduces effective capacity by 5%, increasing stress risk.
Power Factor Penalty	0.15	$\text{clip}(0.9 - \text{power_factor}, 0, 1)$	Reactive power burden. Low power factor means the transformer carries more apparent current for the same real power — increasing copper losses and heating.
Thermal Soak	0.10	Exponential moving sum of hours above 80% load with time constant 4 hours	Accumulated thermal stress from sustained heavy loading. A transformer that has been above 80% for several hours retains heat even after load drops.
EV Load Risk	0.10	$\text{ev_density} \times \text{evening_peak_multiplier}(\text{hour})$	Risk from EV charging demand concentrating in the 18:00-22:00 evening peak. Higher EV density in the area increases this signal during evening hours.
Transformer Age	0.10	$\log(1 + \text{age_years}) / \log(1 + 25)$	Aging degradation factor. Older transformers have reduced insulation integrity and higher failure probability under load. Normalised to a 25-year reference.
Calendar Signal	0.05	Peak hour factor x seasonal factor x event multiplier	Composite time-of-use factor capturing peak hours (morning/evening), high-consumption months, and holiday/event periods when demand spikes are expected.
PV Duck Curve	0.05	$\text{clip}((\text{feeder_load} - \text{baseline_load}) / \text{baseline_load}, 0, 1) \times \text{solar_factor}$	Ramp-up stress from solar generation drop-off at sunset. The 'duck curve' describes the rapid increase in conventional generation needed as solar drops.

6.1 GSI Scoring Formula

```
GSI = sum( weight[k] x signal[k] ) x u_tconf x mape_scale x 100
u_tconf = clip( 1 - PINAW, 0, 1 ) [forecast uncertainty correction]
mape_scale = clip( 1 - MAPE, 0, 1 ) [forecast accuracy correction]
```

The two correction factors adjust GSI downward when the demand forecast is uncertain (wide prediction intervals) or inaccurate (high MAPE), reflecting reduced confidence in the load quantile signal.

6.2 GSI Label Thresholds

GSI Range	Label	Recommended Action
0 – 34	Stable	Normal operation. No intervention needed.
35 – 54	Caution	Monitor load trend. Check for upcoming events or peak periods.
55 – 74	Stressed	Prepare load-shedding contingency. Alert grid operations team.
75 – 100	Critical	Immediate grid operator notification. Risk of transformer failure.

7. Machine Learning Models

7.1 XGBoost Theft Detector

The theft detector is an XGBoost gradient-boosted tree binary classifier trained to distinguish theft meters (label=1) from honest meters (label=0). Because electricity theft is rare in practice, the dataset is highly imbalanced; the model compensates with a dynamically computed **scale_pos_weight**.

Parameter	Value	Purpose
n_estimators	500	Maximum number of boosting rounds
max_depth	6	Maximum tree depth — controls model complexity
learning_rate	0.05	Step size shrinkage to prevent overfitting
subsample	0.8	Fraction of rows sampled per tree — reduces variance
colsample_bytree	0.8	Fraction of features sampled per tree
scale_pos_weight	~19x	Computed as $\text{count}(\text{negative}) / \text{count}(\text{positive})$ — corrects class imbalance
eval_metric	aucpr	Optimises PR-AUC — more appropriate than ROC-AUC for imbalanced data
early_stopping	30 rounds	Stops training when validation PR-AUC stops improving
random_state	42	Reproducibility seed

7.1.1 Input Features (14 total)

#	Feature Name	Source	Description
1	dtw_divergence	DTW clustering	Euclidean distance of meter profile from cluster centroid
2	voltage_stability	Measured voltage	Std dev of 3-phase voltages normalised by 0.15
3	billing_ratio	Billing data	$ 1 - \text{billed}/\text{actual} $ consumption ratio
4	entropy	kWh series	1 - normalised Shannon entropy of 20-bin histogram
5	night_load_anomaly	kWh + timestamps	Night-to-day consumption ratio anomaly
6	dt_balance_error	DT aggregates	Feeder vs. meter sum energy balance error
7	repeat_anomaly	kWh rolling windows	Fraction of windows with Z-score outlier > 3.0
8	power_factor_mean	Measured	Mean power factor across all readings for this meter
9	kwh_mean_7d	Last 672 readings	Mean kWh over the most recent 7-day window
10	kwh_std_7d	Last 672 readings	Std deviation of kWh over the most recent 7-day window

#	Feature Name	Source	Description
1 1	kwh_trend_slope	Linear regression	Slope of linear fit to the full kWh time series (polyfit)
1 2	hour_of_day	Last timestamp	Hour (0-23) of the final reading for temporal context
1 3	day_of_week	Last timestamp	Day of week (0=Monday, 6=Sunday)
1 4	month	Last timestamp	Calendar month (1-12) for seasonal context

7.2 Bi-LSTM Demand Forecaster

The demand forecaster is a bidirectional LSTM (Long Short-Term Memory) neural network that predicts the distribution of future kWh demand at the transformer level. It outputs **two quantiles** — Q5 and Q95 — forming a prediction interval used by the GSI load quantile signal and displayed in the frontend demand chart.

Parameter	Value	Purpose
Architecture	Bidirectional LSTM	Captures both forward and backward temporal dependencies
hidden_size	128	LSTM hidden state dimension per direction (256 total)
num_layers	2	Stacked LSTM layers with dropout between them
dropout	0.2	20% dropout between layers to reduce overfitting
quantiles	[0.05, 0.95]	Q5 and Q95 output heads — defines the prediction interval
lookback_window	24 steps	Hours of history fed into each sequence (24-hour lookback)
epochs	100	Maximum training epochs
batch_size	64	Mini-batch size for stochastic gradient descent
learning_rate	0.001	Adam optimiser initial learning rate
patience	10	Early stopping: halt if validation loss doesn't improve for 10 epochs
Loss function	Quantile loss	Pinball loss computed separately for Q5 and Q95 then summed
Scheduler	ReduceLROnPlateau	Halves learning rate when validation loss plateaus (patience=5)
Normalisation	StandardScaler	All 13 input features scaled to zero mean, unit variance

Parameter	Value	Purpose
Saved artefacts	2 files	models/demand_forecaster.pt (weights) + models/demand_scaler.pkl (scaler)

7.2.1 LSTM Input Features (13 total)

Feature	Description
kwh	Aggregated kWh across all meters under this DT (target variable for training)
temperature_c	Ambient temperature in Celsius — diurnal 24-32 degree C sinusoidal pattern
power_factor	Mean power factor of all meters on this DT
hour_sin	$\sin(2\pi \cdot \text{hour}/24)$ — cyclical encoding of hour of day
hour_cos	$\cos(2\pi \cdot \text{hour}/24)$ — cyclical encoding of hour of day
day_sin	$\sin(2\pi \cdot \text{dayofweek}/7)$ — cyclical encoding of day of week
day_cos	$\cos(2\pi \cdot \text{dayofweek}/7)$ — cyclical encoding of day of week
month_sin	$\sin(2\pi \cdot \text{month}/12)$ — cyclical encoding of calendar month
month_cos	$\cos(2\pi \cdot \text{month}/12)$ — cyclical encoding of calendar month
is_weekend	1.0 if Saturday or Sunday, else 0.0
is_holiday	1.0 if the date is a recognised Indian public holiday, else 0.0
ev_density	Electric vehicle penetration proxy — currently 0.1 (configurable)
solar_irradiance	Solar irradiance W/m^2 — diurnal sinusoidal profile peaking at noon

8. GSS Composite Scoring — Grid Sentinel Score

The GSS (Grid Sentinel Score) is a single composite metric that captures overall system performance across theft detection quality, forecast accuracy, economic efficiency, robustness, calibration, energy balance, and detection speed. Two variants are computed: **GSS Core** (4 components) and **GSS Final** (all 7 components).

8.1 Component Sub-scores

Sub-score	Formula	Measures
S_CASS	$MCC \times (1 - FPR)$	Theft detection quality: balanced accuracy adjusted for false alarm rate
S_GSI	$(1 - MAPE) \times PICP \times (1 - PINAW)$	Forecast quality: accuracy x interval coverage x interval sharpness
S_Econ	$1 - cost / baseline_cost$	Economic efficiency: savings vs. a baseline of no detection
S_Robust	$1 - F1_clean - F1_noisy / F1_clean$	Stability under Gaussian noise injection (std=0.05)
S_Delay	$\exp(-latency_days / 7)$	Detection speed: exponential decay from theft start to detection
S_Energy	$1 - meter_kWh_total - feeder_kWh_total / feeder_kWh_total$	Energy accounting consistency across meters and feeders
S_Calib	$1 - ECE$	Probability calibration: reliability of predicted theft probabilities

8.2 Aggregation Formulas

$$GSS_Core = w_CASS(0.35) \times S_CASS + w_GSI(0.20) \times S_GSI \\ + w_Econ(0.30) \times S_Econ + w_Robust(0.15) \times S_Robust$$

$$GSS_Final = w_CASS(0.30) \times S_CASS + w_GSI(0.15) \times S_GSI \\ + w_Econ(0.25) \times S_Econ + w_Robust(0.10) \times S_Robust \\ + w_Delay(0.08) \times S_Delay + w_Energy(0.07) \times S_Energy \\ + w_Calib(0.05) \times S_Calib$$

Pareto Constraint Penalty: If any of $FPR > 0.02$, $Recall < 0.85$, or $MAPE > 0.07$ is violated, both GSS Core and GSS Final are multiplied by 0.5. This ensures the composite score cannot be high when fundamental operational constraints are breached.

9. Economic Model

GridSentinel quantifies the economic impact of its detection decisions in Indian Rupees. The model distinguishes between the cost of false investigations (false positives) and the cost of missed thefts (false negatives).

Parameter	Value (INR)	Description
FP investigation cost	Rs. 8,500	Cost per field team dispatch to investigate a flagged meter that turns out to be honest
FN monthly theft loss	Rs. 5,500	Monthly revenue loss per undetected theft case (stolen energy + administrative cost)
Default theft duration	3 months	Assumed duration a theft goes undetected if flagged as FN
Total FN cost per case	Rs. 16,500	5,500 x 3 months — total loss for one missed theft
Baseline cost (no detection)	Variable	Cost assuming ALL theft cases remain undetected (FN_count + TP_count) x 16,500
System cost	Variable	FP_count x 8,500 + FN_count x 16,500
Economic benefit (S_Econ)	0.0 to 1.0	$1 - \text{system_cost} / \text{baseline_cost}$ — fraction of potential loss avoided

9.1 Frontend Economic Summary Panel

- **Critical Meters** (CASS > 80): Count of meters flagged for immediate intervention.
- **Watch List** (CASS 50–80): Count of meters under monitoring.
- **Estimated Revenue Protection**: $\text{critical_count} \times 5,500 \times 3$ — revenue recoverable if all critical meters are confirmed theft.
- **Investigation Cost**: $\text{watch_count} \times 8,500$ — estimated cost of dispatching teams to all watch-list meters.
- **Net Benefit**: Revenue protection minus investigation cost.

10. Frontend UI — Tabs & Components

The GridSentinel frontend is a React 19 single-page application served by Vite on `http://localhost:5173` during development. All API calls use the `/api` prefix proxied by Vite to the FastAPI backend on port 8000. Authentication uses JWT bearer tokens stored in `localStorage`.

10.1 Login Page

- **Username field:** Accepts the BESCOM operator username.
- **Password field:** Masked input for the bcrypt-hashed password.
- **Default credentials:** username = `admin`, password = `bescom2026` (seeded at startup).
- On successful login, a JWT bearer token (valid 60 minutes) is stored in `localStorage` as `gs_token`.
- 401 responses from any subsequent API call trigger automatic logout and redirect to login.

10.2 Main Dashboard — Left Panel

Live Score Feed (Server-Sent Events, updates every 10 seconds):

Element	Description
meter_id	Meter identifier string (e.g. MTR_000042)
risk score	CASS score 0-100 with small random noise (+/- 2) for live animation
status badge	Colour-coded: normal (green), moderate (yellow), high (orange), critical (red)

Anomalous Cluster Feed Table (top-8 meters by CASS score):

Column	Description
Feeder	Meter ID — click to load DTW chart and SHAP breakdown for this meter
Risk Score	CASS score (0-100). Colour coded by severity.
Logic	'DTW + CASS' when risk $\geq$ 65, else 'CASS' only
Status	Text label: normal / moderate / high / critical

10.3 DTW Divergence Chart (Centre Panel)

- **Triggered by:** clicking any row in the Anomalous Cluster Feed, or querying a meter ID in the AI chat.
- **X-axis:** Hour of day, 0 to 23.
- **Y-axis:** Average kWh consumption.
- **Peer Average line:** Mean hourly consumption of all other meters connected to the same distribution transformer.
- **Target line:** Hourly profile of the selected meter.
- **Interpretation:** Large divergence between Target and Peer Average at specific hours indicates suspicious behaviour. A thief using a bypass shows lower consumption than peers at high-usage times.

10.4 SHAP Feature Importance Panel

- **Triggered by:** same selection as DTW chart.

- Displays a horizontal bar chart of all 7 CASS signal contributions: *value x weight*.
- Bars are sorted by absolute contribution — the most influential signal appears at the top.
- Contribution values are computed using the actual meter's signals from the stored meter_df.
- **DTW Divergence** and **DT Balance Error**: computed against population mean and full DT aggregate respectively.
- **Purpose**: Explains to the operator exactly which signals drove the suspicion score, enabling targeted field investigation.

10.5 Prediction / Demand Forecast Tab

- **Transformer selector**: Dropdown listing all DT IDs loaded from the CSV (DT_1 through DT_N).
- **GSI Score badge**: Large coloured number showing the current Grid Stress Index (0-100) for the selected transformer.
- **Recommended Action**: Text derived from GSI label — e.g. 'Immediate Grid Operator Notification' for Critical.
- **Stress Factors table**: Lists each contributing GSI signal with its computed value.

Demand Chart — 4 lines per hour (0-23):

Line	Colour	Description
Actual	Blue	Measured feeder kWh averaged by hour from the stored dt_df
Predicted	Green	Q50 median forecast from Bi-LSTM (middle quantile if 3 quantiles, Q95 if 2)
Q5 Lower	Orange	5th percentile forecast — lower bound of the prediction interval
Q95 Upper	Red	95th percentile forecast — upper bound of the prediction interval

10.6 AI Chat Panel

- **Input**: Natural language text field — accepts free-form queries in English.
- **Backend**: POST /api/chat — tries Ollama (llama3.2) first; falls back to keyword matching if Ollama unavailable.
- **System prompt**: GridSentinel is briefed as a BESCOM grid expert; responses are limited to 2-3 sentences with a recommended action.
- **Meter query** (e.g. 'MTR_000001 anomaly'): Returns CASS score + automatically triggers DTW chart for that meter.
- **Transformer query** (e.g. 'DT_3 stress'): Returns GSI commentary + triggers demand forecast panel for that DT.
- **Theft/anomaly keywords**: Returns top-3 highest CASS meters from live store.
- **Grid/GSI keywords**: Explains the 8 GSI signal composition.

10.7 Audit Log Tab

Column	Description
Timestamp	UTC datetime of the action
Action	Action type: anomaly_detail, demand_detail, etc.
Target ID	The meter or transformer ID that was queried
Target Type	'meter' or 'transformer'

Column	Description
Details	Human-readable description of the operation

The audit log auto-refreshes every 30 seconds. It records every call to `/api/anomaly/{id}` and `/api/demand/{id}`, providing a full operator activity trail. Requires authentication.

11. API Endpoints Reference

The FastAPI backend exposes the following endpoints on port 8000. All `/api/*` endpoints use relative paths proxied by Vite during development. CORS is pre-configured for `localhost:5173`.

Method	Endpoint	Auth	Description
GET	<code>/health</code>	No	Returns API version and status string
POST	<code>/api/auth/login</code>	No	Verifies credentials, returns JWT bearer token + username
POST	<code>/api/auth/register</code>	No	Creates a new user account with bcrypt-hashed password
POST	<code>/api/auth/token</code>	No	OAuth2 alias for <code>/api/auth/login</code>
GET	<code>/api/anomaly/clusters</code>	No	Returns top-N meters ranked by CASS score. Default N=8.
GET	<code>/api/anomaly/{meter_id}</code>	No	Returns hourly DTW series (peer vs target) and CASS score for one meter
GET	<code>/api/shap/{meter_id}</code>	No	Returns 7 CASS signal values and weighted contributions for one meter
GET	<code>/api/demand/{dt_id}</code>	No	Returns hourly actual vs forecast demand, GSI score, action and stress factors for one DT
GET	<code>/api/economic/summary</code>	No	Returns critical/watch counts and INR economic KPIs
GET	<code>/api/map/zones</code>	No	Returns per-DT risk summary for the zone map view
GET	<code>/api/stream/scores</code>	No	Server-Sent Events stream: top-3 meter scores + GSI every 10 seconds
POST	<code>/api/chat</code>	No	Natural language query — Ollama LLM with keyword fallback
GET	<code>/api/audit/log</code>	YES	Returns last 50 audit log entries. JWT required.
GET	<code>/api/report/summary</code>	No	Generates and streams a PDF intelligence report
POST	<code>/score/meter</code>	No	Scores a single meter from supplied records via CASS formula
POST	<code>/score/transformer</code>	No	Scores a single transformer from supplied DT records via GSI formula
POST	<code>/evaluate</code>	No	Runs the full 18-step pipeline on supplied meter and DT records

12. Classification Metrics Explained

The following metrics are computed by `evaluate_classification()` after the `TheftDetector` predicts on the full feature matrix. They appear in the pipeline summary output and are used to compute `S_CASS` for the GSS score.

Metric	Full Name	Formula	Interpretation
Precision	Precision	$TP / (TP + FP)$	Of all meters flagged as theft, what fraction truly are. High precision = few wasted field investigations.
Recall	Recall / Sensitivity	$TP / (TP + FN)$	Of all actual theft meters, what fraction were caught. High recall = fewer missed thefts. Constraint: ≥ 0.85
F1	F1 Score	$2 \times \frac{\text{Precision} \times \text{Recall}}{(\text{Precision} + \text{Recall})}$	Harmonic mean — balances precision and recall. Primary metric for imbalanced datasets.
FPR	False Positive Rate	$FP / (FP + TN)$	Fraction of honest meters incorrectly flagged. Constraint: ≤ 0.02 (at most 2% false alarms).
MCC	Matthews Correlation Coefficient	$(TP \cdot TN - FP \cdot FN) / \sqrt{...}$	Balanced metric for imbalanced classes. Ranges -1 to +1. Only metric that uses all 4 confusion matrix cells equally.
ROC-AUC	Area Under ROC Curve	Integral of TPR vs FPR curve	Probability that the model ranks a random theft meter above a random honest meter. 0.5 = random, 1.0 = perfect.
PR-AUC	Area Under PR Curve	Integral of Precision vs Recall curve	More informative than ROC-AUC for heavily imbalanced data (rare theft). Used as XGBoost <code>eval_metric</code> .
ECE	Expected Calibration Error	Mean $ \text{accuracy} - \text{confidence} $ per bin	Measures how well predicted probabilities match true frequencies in 10 reliability bins. Used to compute <code>S_Calib</code> .

13. Forecast Metrics Explained

Forecast metrics are computed by `evaluate_forecast()` comparing Bi-LSTM Q5/Q95 predictions against actual feeder kWh values. They feed into `S_GSI` for the GSS score and are reported in the pipeline summary. Constraint: MAPE $\leq$ 0.07.

Metric	Full Name	Formula	Interpretation
MAPE	Mean Absolute Percentage Error	$\text{mean}(\text{actual} - \text{pred} / \text{actual}) \times 100\%$	Average percentage forecast error. 7% MAPE means forecasts are off by 7% on average. Constraint: $\leq 7\%$.
RMSE	Root Mean Squared Error	$\sqrt{\text{mean}((\text{actual} - \text{pred})^2)}$	Penalises large errors more than MAPE. Measured in kWh. Useful for detecting infrequent large forecast failures.
PICP	Prediction Interval Coverage Probability	$\text{mean}(Q5 \leq \text{actual} \leq Q95)$	Fraction of actual values that fall within the Q5-Q95 prediction band. Ideal value: 0.90 (90% coverage).
PINAW	Prediction Interval Normalised Average Width	$\text{mean}(Q95 - Q5) / \text{range}(\text{actual})$	Width of the prediction interval relative to the data range. Lower = sharper (more confident) intervals. Trade-off vs PICP.

13.1 Pareto Constraint Summary

Constraint	Threshold	Metric	Consequence if Violated
Max FPR	≤ 0.02	FPR	GSS Core and Final multiplied by 0.5 — system penalised for excessive false alarms
Min Recall	≥ 0.85	Recall	GSS Core and Final multiplied by 0.5 — system penalised for missing too many thefts
Max MAPE	≤ 0.07	MAPE	GSS Core and Final multiplied by 0.5 — system penalised for poor demand forecasting

This document was automatically generated by GridSentinel AI v2.0 on 07 May 2026 at 17:35. All technical specifications reflect the implementation in the GridSentinel codebase as reviewed and documented during the BESCOM Hackathon.